Calculus II

Tutorial #10 21-25 Apr 2025

Topics:

- Exact differential equations
- Numerical solutions to differential equations

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. There is a unique potential function for an exact differential equation
$$M(x, y)\, dx + N(x, y)\, dy = 0.$$
2. The solution to an exact differential equation is called a potential function.
3. Be able to apply Euler's method to approximate the solution to an IVP at a point near the initial value x_0.

• **Problems**.

4. Determine whether the following differential equations are exact
   (a) $[\sin(xy) + xy\cos(xy) + 2x]\ dx + [x^2\cos(xy) + 2y]\ dy = 0.$
   (b) $(y^2 + \cos x)\, dx + (2xy + \sin y)\, dy = 0.$
5. Solve the differential equation
$$2xy\, dx + (x^2 + 1)\, dy = 0$$
6. Solve the differential equation
$$(y^2 - 2x)\, dx + 2xy\, dy = 0$$
7. Solve the initial-value problem
$$(3x^2 \ln x + x^2 - y)\, dx - x\, dy = 0, \quad y(1) = 5$$
8. Solve the initial-value problem
$$2x^2y' + 4xy = 3\sin x, \quad y(2\pi) = 0$$

9. Use Euler's method with the specified step size to determine the solution to the given IVP at the specific point.

$$y' = -x^2 y, \quad y(0) = 1, \quad h = 0.2, \quad y(1).$$

10. Use Euler's method with the specified step size to determine the solution to the given IVP at the specific point.

$$y' = 4y - 1, \quad y(0) = 1, \quad h = 0.05, \quad y(0.5).$$

END

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Answers to selected problems

1. False 2. False 3. True
4. (a) Yes (b) Yes
5. $(x^2 + 1)y = C$
6. $xy^2 - x^2 = C$
7. $y(x) = \dfrac{x^3 \ln x + 5}{x}$
8. $y(x) = \dfrac{3 - 3\cos x}{2x^2}$
9. $y_5 = 0.777$
10. $y_{10} = 4.894$.